CHECKPOINT 3

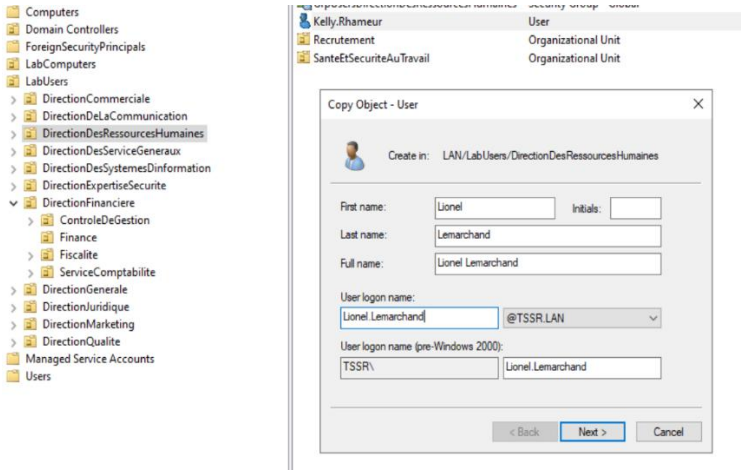
Formulaire réponses

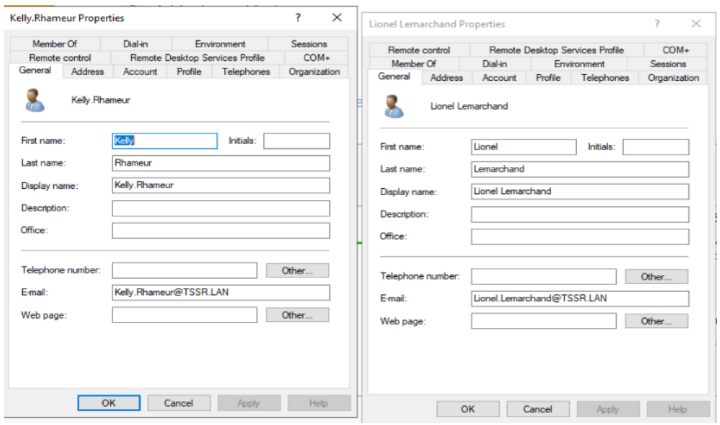
Exercice 1

Partie 1

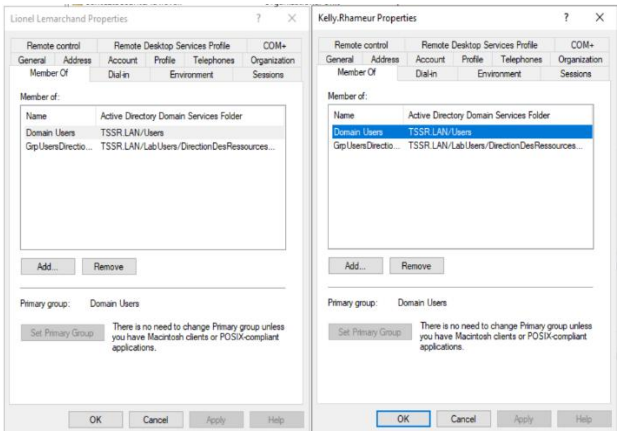
Q.1.1.1

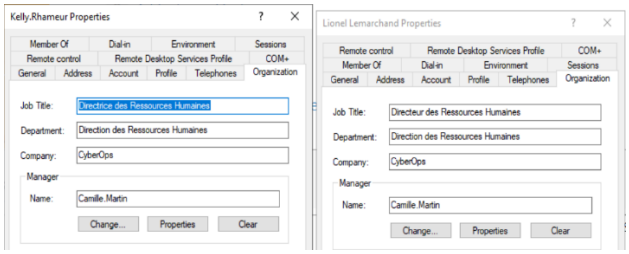
Copie(s) d'écran montrant que Lionel Lemarchand a les mêmes attributs de société que Kelly Rhameur.





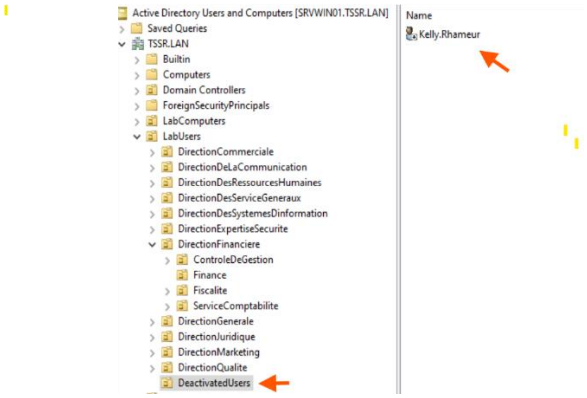
Copie(s) d’écran montrant le changement coté management.





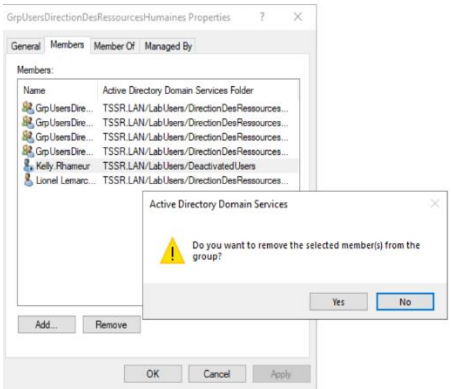
Q.1.1.2

Copie d’écran de l’OU DeactivatedUsers.



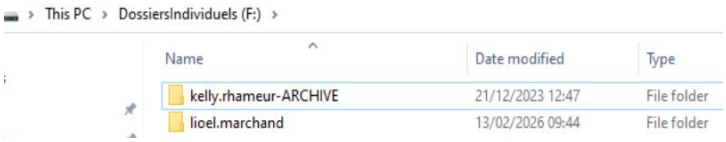
Q.1.1.3

Copie d’écran de l’ancien groupe dans lequel était Kelly Rhameur.



Q.1.1.4

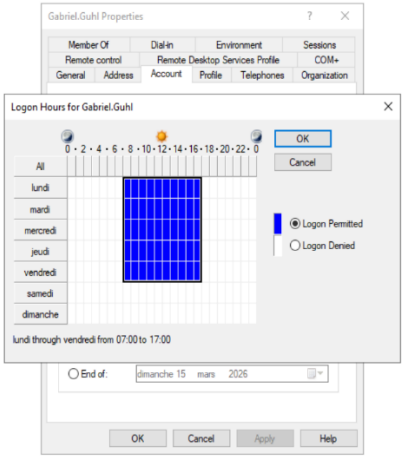
Copie d’écran des dossiers individuels demandés.



Partie 2

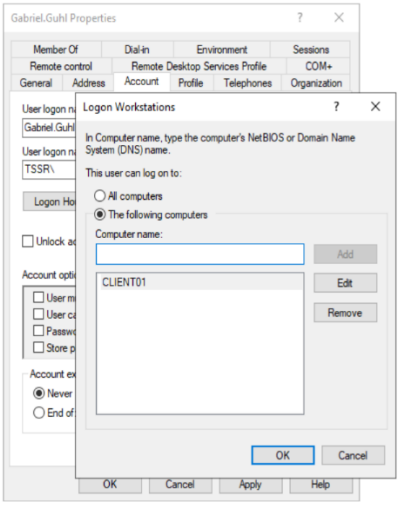
Q.1.2.1

Copies d’écran montrant le paramétrage de la restriction de connexion horaire.



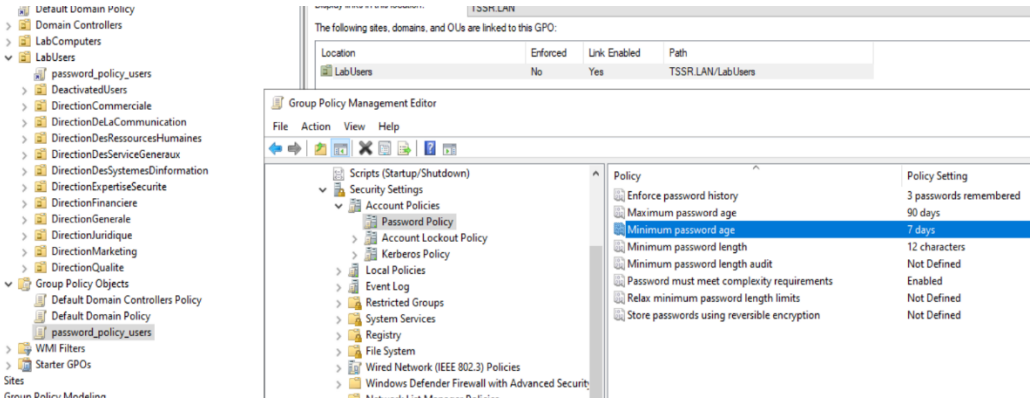
Q.1.2.2

Copie d’écran montrant le paramétrage de la restriction de la connexion machine.



Q.1.2.3

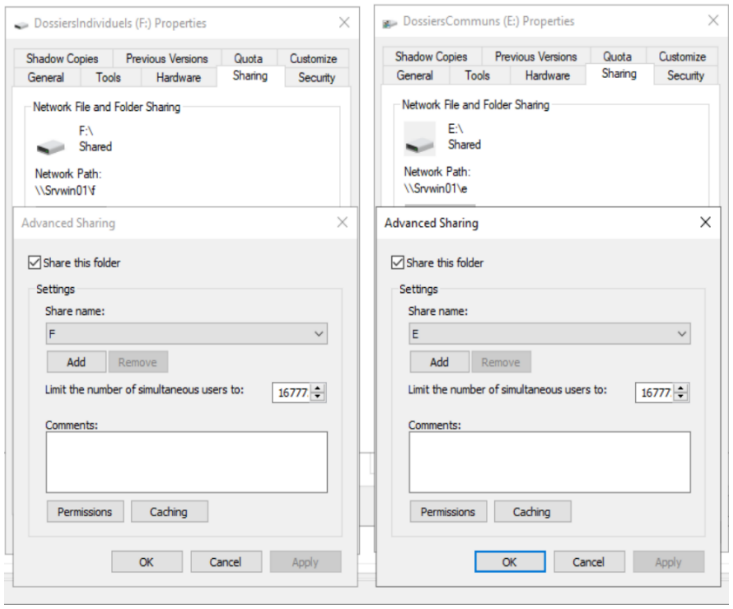
Copies d'écran de la stratégie de mot de passe.

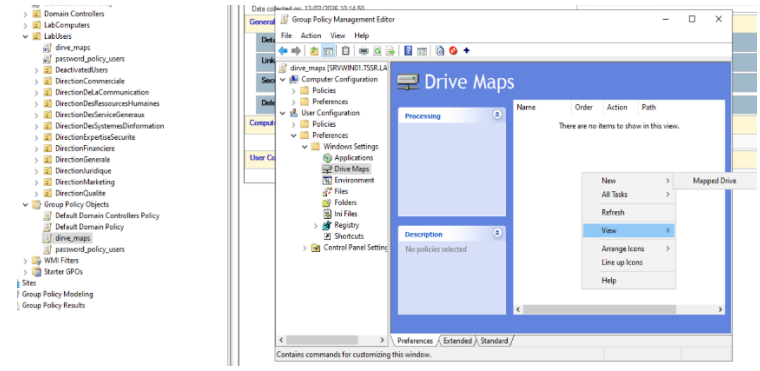


Partie 3

Q.1.3.1

Copies d'écran de la GPO de montage de lecteurs.





Exercice 2

Partie 1

Q.2.1.1

Copie d'écran de la création de compte.

```
root@SRVLX01:~# useradd -m -s /bin/bash Safiullah_
```

```
root@SRVLX01:~# cat /etc/passwd | grep Safiullah
Safiullah:x:1001:1001::/home/Safiullah:/bin/bash
root@SRVLX01:~# _
```

Q.2.1.2

Copie d'écran du paramétrage du compte.

```
root@SRVLX01:~# passwd Safiullah
Nouveau mot de passe :
Retapez le nouveau mot de passe :
passwd: password updated successfully
root@SRVLX01:~# usermod -aG sudo Safiullah
root@SRVLX01:~# _
```

Partie 2

Q.2.2.1

Copie d'écran du paramétrage de l'accès distant.

```
#LoginGraceTime 2m
PermitRootLogin no
```

```
PS C:\WINDOWS\system32> ssh root@192.168.1.200
root@192.168.1.200's password:
Permission denied, please try again.
root@192.168.1.200's password:
```

Q.2.2.2

Copie d'écran du paramétrage distant avec le compte personnel.

```
AllowUsers safiullah
```

```
PS C:\WINDOWS\system32> ssh Safiullah@192.168.1.200
Safiullah@192.168.1.200's password:
Linux SRVLX01 5.10.0-20-amd64 #1 SMP Debian 5.10.158-2 (2022-12-13) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Safiullah@SRVLX01:~$
```

Q.2.2.3

Copies d'écran du paramétrage de l'authentification.

```
root@SRVLX01:~# ssh-keygen -t rsa -b 4096
Generating public/private rsa key pair.
Enter file in which to save the key (/root/.ssh/id_rsa):
Created directory '/root/.ssh'.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /root/.ssh/id_rsa
Your public key has been saved in /root/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:cwukAQW8O/55Btw3xNkOpbh8wxaq5clA10rtmdBspiw root@SRVLX01
The key's randomart image is:
+---[RSA 4096]----+
|    .oo.         |
|     ..     .    |
|      .. .B =    |
|     . .+= # .   |
|      +.*S%.*    |
|     o E X+@..   |
|    . . O =.o    |
|     . ..*       |
|      .oo        |
+----[SHA256]-----+
root@SRVLX01:~#
```

```
Include /etc/ssh/sshd_config.d/*.conf

PermitRootLogin no
AllowUsers Safiullah

#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin prohibit-password
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

PubkeyAuthentication yes
```

Partie 3

Q.2.3.1

Copie d'écran montrant les systèmes de fichiers montés.

```
root@SRVLX01:~# df -hT
Sys. de fichiers         Type      Taille Utilisé Dispo Uti% Monté sur
udev                     devtmpfs    470M       0  470M   0% /dev
tmpfs                    tmpfs        98M    620K   98M   1% /run
/dev/mapper/cp3--vg-root ext4        2,7G    1,5G  1,1G  60% /
tmpfs                    tmpfs       489M      16K  489M   1% /dev/shm
tmpfs                    tmpfs       5,0M       0  5,0M   0% /run/lock
/dev/md0p1               ext2        471M      49M  398M  11% /boot
tmpfs                    tmpfs        98M       0   98M   0% /run/user/0
tmpfs                    tmpfs        98M       0   98M   0% /run/user/1001
root@SRVLX01:~#
```

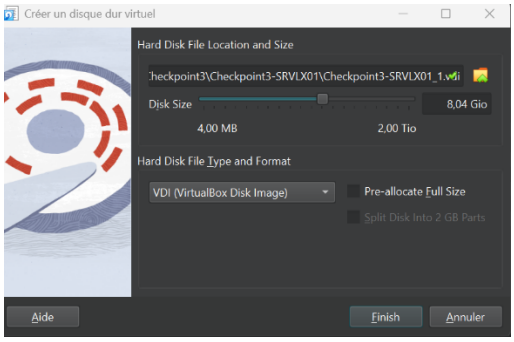
Q.2.3.2

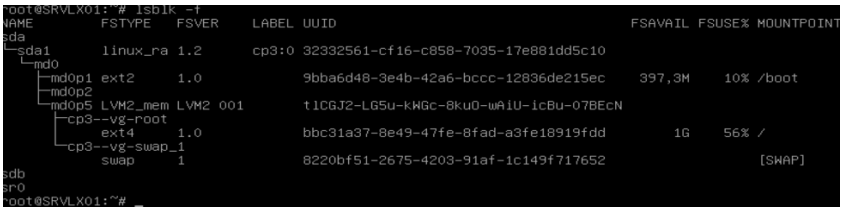
Copie d'écran montrant les systèmes de stockage utilisés.

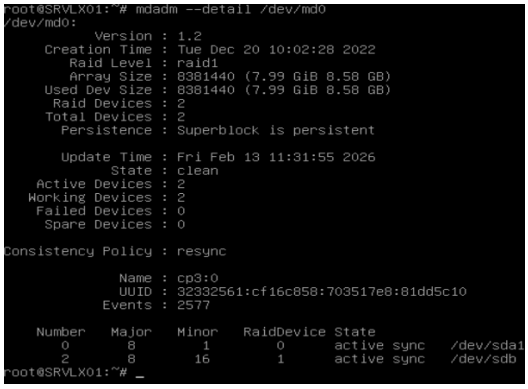
```
root@SRVLX01:~# lsblk
NAME                  MAJ:MIN RM   SIZE RO TYPE  MOUNTPOINT
sda                     8:0    0     8G  0 disk
└sda1                   8:1    0     8G  0 part
  └md0                  9:0    0     8G  0 raid1
    ├md0p1            259:0    0 488,3M  0 part  /boot
    ├md0p2            259:1    0     1K  0 part
    └md0p5            259:2    0   7,5G  0 part
      ├cp3--vg-root   253:0    0   2,8G  0 lvm   /
      └cp3--vg-swap_1 253:1    0   976M  0 lvm   [SWAP]
sr0                    11:0    1  1024M  0 rom
root@SRVLX01:~#
```

Q.2.3.3

Copies d’écran montrant les différentes étapes pour la réparation du volume RAID.

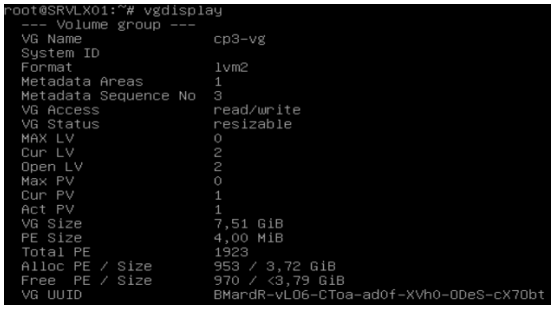


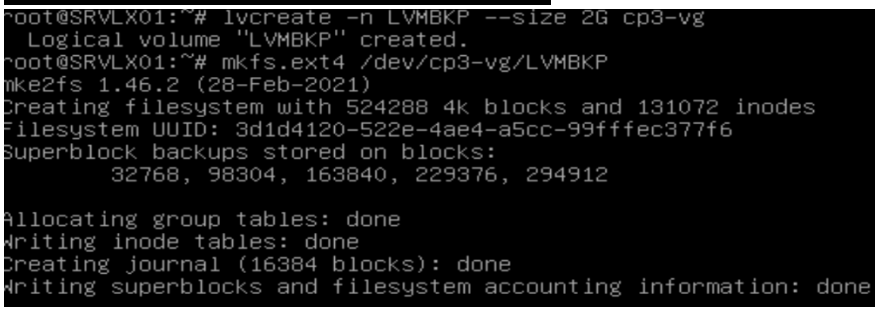




Q.2.3.4

Copies d’écran montrant les différentes étapes de configuration.





```
root@SRVLX01:~# mkdir -p /var/lib/bareos/storage
root@SRVLX01:~# mount /de/cp3-vg/LVMBKP /var/lib/bareos/storage
mount: /var/lib/bareos/storage: le périphérique spécial /de/cp3-vg/LVMBKP n'existe pas
root@SRVLX01:~# mount /dev/cp3-vg/LVMBKP /var/lib/bareos/storage
root@SRVLX01:~# lsblk -l
NAME            MAJ:MIN RM   SIZE RO TYPE  MOUNTPOINT
sda               8:0    0     8G  0 disk
sda1              8:1    0     8G  0 part
sdb               8:16   0     8G  0 disk
md0               9:0    0     8G  0 raid1
sr0              11:0    1  1024M  0 rom
cp3--vg-root   253:0    0   2,8G  0 lvm   /
cp3--vg-swap_1 253:1    0   976M  0 lvm   [SWAP]
cp3--vg-LVMBKP 253:2    0     2G  0 lvm   /var/lib/bareos/storage
md0p1          259:0    0 488,3M  0 part  /boot
md0p2          259:1    0     1K  0 part
md0p5          259:2    0   7,5G  0 part
root@SRVLX01:~# _
```

```
# /etc/fstab: static file system information.
#
# Use 'blkid' to print the universally unique identifier for a
# device; this may be used with UUID= as a more robust way to name devices
# that works even if disks are added and removed. See fstab(5).
#
# systemd generates mount units based on this file, see systemd.mount(5).
# Please run 'systemctl daemon-reload' after making changes here.
#
# <file system> <mount point>   <type>  <options>       <dump>  <pass>
/dev/mapper/cp3--vg-root /               ext4    errors=remount-ro 0       1
# /boot was on /dev/md0p1 during installation
UUID=9bba6d48-3e4b-42a6-bccc-12836de215ec /boot           ext2    defaults        0       2
/dev/mapper/cp3--vg-swap_1 none            swap    sw              0       0
/dev/sr0        /media/cdrom0   udf,iso9660 user,noauto     0       0
/dev/cp3-vg/LVMBKP /var/lib/bareos/storage ext4         defaults        0       0
```

Q.2.3.5

Copie d'écran montrant l'espace disponible.

```
root@SRVLX01:~# vgs
  VG     #PV #LV #SN Attr   VSize VFree
  cp3-vg   1   3   0 wz--n- 7,51g <1,79g
```

Partie 4

Q.2.4.1

Bareos-director (dir) : Il gère les configurations, planifie et supervise les sauvegardes, s'occupe du stockage, coordonne tous les processus de sauvegarde, restauration et stock les métadonnées.
Bareos-Storage Daemon (sd) : Il gère le stockage physique des données, et lit les données sur les supports de stockage communique avec le director et les file daemons.
Bareos-file Daemon (fd) : il lit et envoie les fichiers à sauvegarder, retaure les fichiers si il faut faire une restauration et communique avec director.

Partie 5

Q.2.5.1

```
root@SRVLX01:~# nft list ruleset
table inet inet_filter_table {
        chain in_chain {
                type filter hook input priority filter; policy drop;
                ct state established,related accept
                ct state invalid drop
                iifname "lo" accept
                tcp dport 22 accept
                ip protocol icmp accept
                ip6 nexthdr ipv6-icmp accept
        }
```

Q.2.5.2

Les connexions établies et reliées

Trafic sur l'interface locale (lo)

TCP port 22 ssh

Ip protocol les pings ICMP

IP6 ipv6-icmp

Q.2.5.3

1: Paquets invalides: (ct state invalid drop)

2 : Toutes les nouvelles connexions non explicitement autorisées (politique DROP par défaut)

Q.2.5.4

J’ai modifié le fichier nftables/config.nft et j’ai ajouté le règles tcp dport 9101-9103 ct new accept.

```
flush ruleset

table inet inet_filter_table {
        chain in_chain {
                type filter hook input priority filter; policy drop;
                ct state established, related accept
                ct state invalid drop
                iifname lo accept
                tcp dport ssh accept
                ip protocol icmp accept
                ip6 nexthdr icmpv6 accept
                tcp dport { 9101-9103 } ct state new accept
        }
```

Partie 6

Q.2.6.1

```
Safiullah@SRVLX01:~$ grep "Failed password" /var/log/auth.log | tail -n 10
grep: /var/log/auth.log: Permission non accordée
Safiullah@SRVLX01:~$ su root
Mot de passe :
root@SRVLX01:/home/Safiullah# grep "Failed password" /var/log/auth.log | tail -n 10
Feb 13 10:53:18 SRVLX01 sshd[1024]: Failed password for invalid user root from 192.168.1.33 port 54900 ssh2
Feb 13 10:53:34 SRVLX01 sshd[1027]: Failed password for invalid user root from 192.168.1.33 port 63395 ssh2
root@SRVLX01:/home/Safiullah# journalctl -u ssh -g "Failed password" -n 10
-- Journal begins at Tue 2022-12-20 10:13:18 CET, ends at Fri 2026-02-13 13:10:00 CET. --
févr. 13 10:53:18 SRVLX01 sshd[1024]: Failed password for invalid user root from 192.168.1.33 port 54900 ssh2
févr. 13 10:53:34 SRVLX01 sshd[1027]: Failed password for invalid user root from 192.168.1.33 port 63395 ssh2
root@SRVLX01:/home/Safiullah#
```